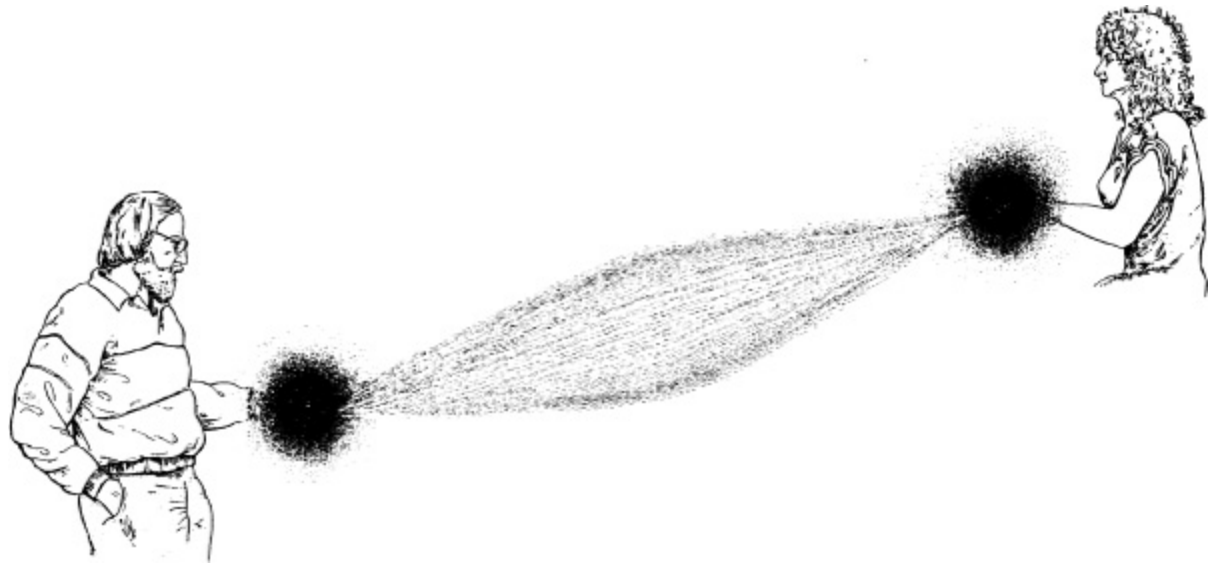


14.11 As Carolee and I try to convert a wormhole into a time machine by the method of Figure 14.7, electromagnetic vacuum fluctuations zoom between the two mouths and through the wormhole, piling up on themselves and creating a beam of huge fluctuational energy.

Therefore, Kim and I reasoned, the circulating electromagnetic vacuum fluctuations must stop evolving with time, that is, must stop growing, 10^{-43} second before the wormhole becomes a time machine; the laws of quantum gravity must cut off the fluctuations' growth. And the quantum gravity laws will let the fluctuations continue their evolution again only 10^{-43} second after the time machine is born, which means after they have begun to die out. In between these times, there is no *time* and there is no evolution (Figure 14.12). The crucial issue, then, was *just how intense has the beam of circulating fluctuations become when quantum gravity cuts off their growth?* Our calculations were clear and unequivocal: The beam, when it stops growing, is far too weak to damage the wormhole, and therefore, in the words of our manuscript, it seemed likely that "vacuum fluctuations cannot prevent the formation of or existence of closed time-like curves." *(Closed time-like curves* is physicists' jargon for "time machines"; having been burned by the press, I had stopped using the phrase "time machines" in my papers; and the press, unfamiliar with physicists' jargon, was now unaware of the new time machine results I was publishing.)